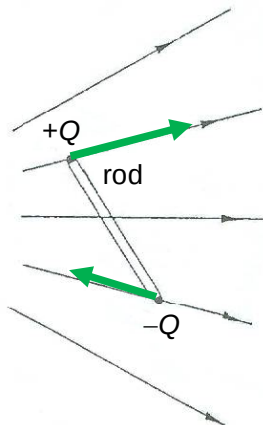


- 18 An insulated rod with equal and opposite charges at its ends is placed in a non-uniform electric field with field lines as shown below.



The rod experiences

- A** a resultant force in the plane of the paper and a torque.
- B** a resultant force in the plane of the paper but no torque.
- C** a resultant force normal to the plane of the paper and a torque.
- D** a resultant force normal to the plane of the paper but no torque.

The resultant force can be seen from the diagram that it lies in the plane of the paper. The forces will produce a clockwise torque and cause the rod to rotate.

19 Four point charges are at the corners of a square JKLM. The point charges at J and K have a